



Module 3 challenge

1. Fill in the blank: Data professionals use data _____ to handle issues related to internal and external data flows while ensuring data assets are formally managed.

1 / 1 point

- ☐ integrity
- ☒ governance
- ☐ strategy
- ☐ mapping

✓ Correct

2. An analyst receives a dataset in a CSV file named `sales_2024.csv`. This file contains records of product sales for the last fiscal year, including columns like `transaction_id`, `product_sku`, `sale_date`, `quantity_sold`, and `sale_amount`. Which of the following are examples of metadata that might be associated with this dataset? Select all that apply.

1 / 1 point

- ☐ The maximum value in the `sale_amount` column.
- ☒ The date and time the `sales_2024.csv` file was last modified on the server

✓ Correct

- ☒ A data dictionary file that defines each column name (`transaction_id`, `product_sku`, etc.) and specifies its data type (e.g., Text, Date, Integer, Decimal)

✓ Correct

- ☒ The file's size, location, and access permissions

✓ Correct

3. Which of the following statements accurately describe primary and foreign keys in a relational database? Select all that apply.

1 / 1 point

☒ A table can only have one primary key, but it can have multiple foreign keys.

✓ Correct

☒ Foreign keys are used to connect one table to another.

✓ Correct

☐ A foreign key uniquely identifies a record in the table in which it is located.

☒ A primary key references a column in which each value is unique.

✓ Correct

4. A data analyst runs the following query. What do they want to retrieve from the database?

1 / 1 point

```
1 SELECT *
2 FROM CustomerOrders
3 WHERE Country = 'Finland';
```

- ☒ All fields of orders from customers in Finland
- ☐ All fields of the customers who have placed an order
- ☐ All fields of the names of customers living in Finland
- ☐ All fields of orders from customers outside of Finland

✓ Correct

5. What are some benefits of using internal data for an analysis project? Select all that apply.

1 / 1 point

☐ Internal data represents the entire industry.

☒ Internal data is free to access because the company owns it.

✓ Correct

☒ Internal data is likely to be reliable

✓ Correct

☐ Internal data fully represents a universal topic.

6. A data professional at a local zoo is sending an email to all donors who give at least \$25 to the zoo each year. What spreadsheet tool will enable them to display only donors who meet that condition?

1 / 1 point

- ☐ Filter the data to show only donors who have given less than \$25 each year
- ☒ Filter the data to show only donors who have given more than \$25 each year
- ☐ Sort donor names in ascending order
- ☐ Sort donor names in descending order

✓ Correct

7. A data analyst at a retail company uses SQL to explore the data in its customer database. They want to examine the **order_date** field in the **CustomerPurchases** table to understand when customers make purchases. What query should they write to return only this column of information?

1 / 1 point

☐

```
1 SELECT *
2 FROM CustomerPurchases
3 WHERE field = "order_date";
```

☒

```
1 SELECT order_date
2 FROM CustomerPurchases;
```

☐

```
1 SELECT *
2 FROM CustomerPurchases;
```

☐

```
1 SELECT CustomerPurchases
2 FROM order_date;
```

✓ Correct

8. Which SQL statement will return only green cars from the Color column of the Cars database table?

1 / 1 point

☐

```
1 SELECT *  
2 FROM Color = 'Green';
```

☒

```
1 SELECT *  
2 FROM Cars  
3 WHERE Color = 'Green';
```

☐

```
1 SELECT *  
2 WHERE 'Green';
```

☐

```
1 SELECT *  
2 FROM Cars  
3 WHERE 'Green';
```

✓ Correct

9. What type of data initiatives can lead to improvements in public planning and an increase in innovation?

1 / 1 point

- ☐ Data privacy initiatives
- ☐ Data governance initiatives
- ☐ Internal data initiatives
- ☒ Open data initiatives

✓ Correct

10. You are a database administrator for a car dealership. The company is growing, and the number of cars the company needs to keep track of has increased. The limitations of the current database design have become more noticeable. There is a lot more repeated data, such as car makes, models, and VIN numbers. This redundancy is causing issues with data integrity and making queries slow. In the tables below what are the primary and foreign keys?

1 / 1 point

Car Table
car_VIN

Make Table
car_make
car_VIN

Model Table
car_model
car_VIN

- ☐ The primary keys would be car_VIN from the Make and Model Tables and the foreign key would be the car_VIN from the Car Table.
- ☒ The primary key would be car_VIN from the Car Table and the foreign keys would be car_VIN from the Make and Model tables.
- ☐ The primary key would be the car_VIN from the Car Table and the foreign key would be the car_model from the Model Table.
- ☐ The primary key would be the car_VIN from the Make Table and the foreign key would be the car_VIN from the Model Table.

✓ Correct